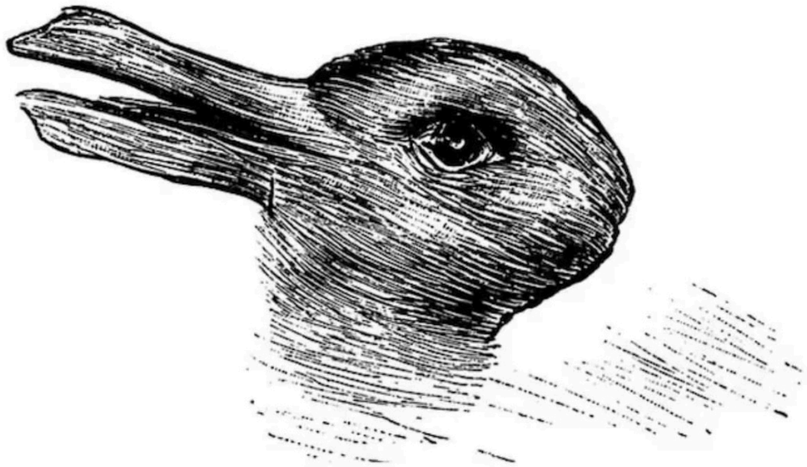
Redundancy of length in physics



Tsampodimos, Apostolos – apostolos.tsampodimos@ftmc.lt

Supervisor: Karčiauskas, Mindaugas

Why modified gravity?

General relativity

classical theory

geometrical description of gravity

?

?

?

Dark energy

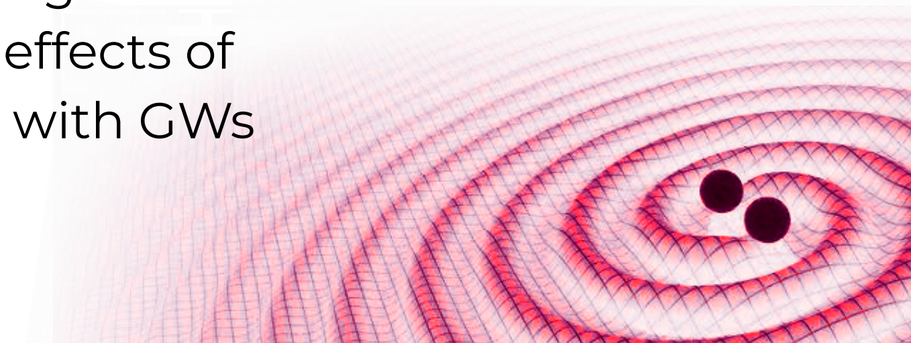
Early Universe

Black holes

Quantum mechanics

Probing non-linear effects of gravity with GWs

Motivation for **modified gravity**



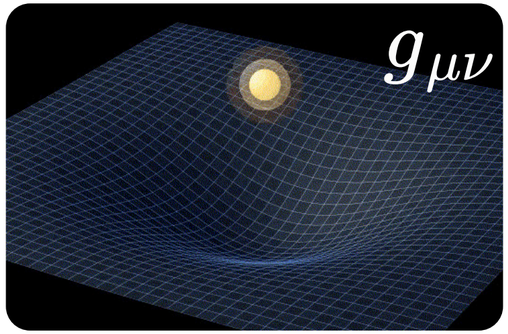
Geometry

$$\mathrm{G}[g_{\mu\nu}] = \mathrm{T}_{\mu\nu}$$

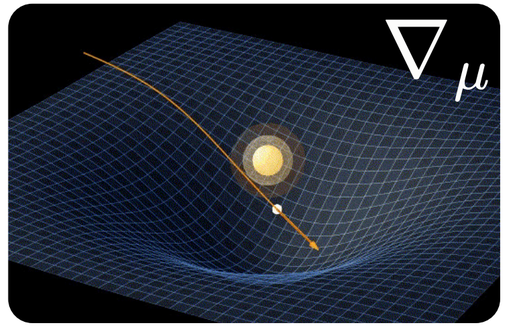
metric

↓

geometry of spacetime



$(\mathcal{M}, \nabla, g)$

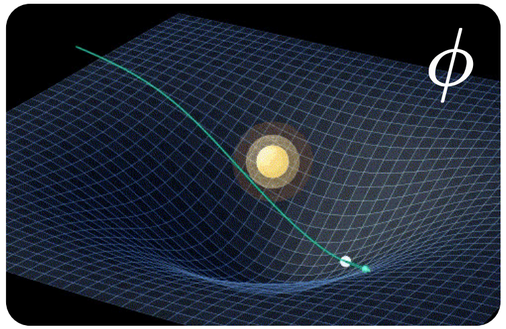


connection

↓

how to move in spacetime

↓ + scale



Integrable Weyl geometry

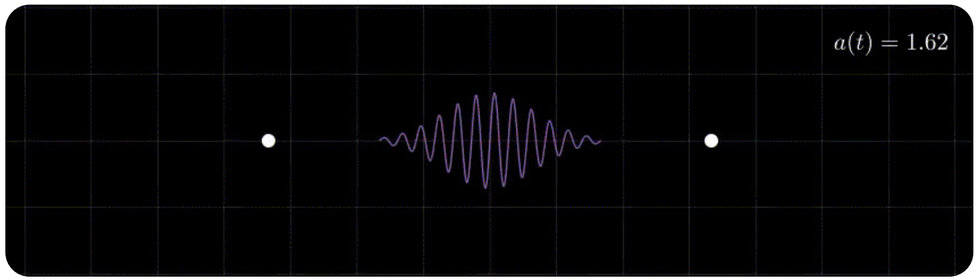
$$(\mathcal{M}, \nabla, g, \phi)$$

We make the choice of scale explicit

$$\mathrm{G}[g_{\mu\nu}] - 2(\nabla_\mu \nabla_\nu \phi - \partial_\mu \phi\, \partial_\nu \phi) + (2\, g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi + g^{\alpha\beta} \partial_\alpha \phi\, \partial_\beta \phi) g_{\mu\nu} = e^{-2\phi}\, \mathrm{T}_{\mu\nu}$$

Interpretation

$$(g^{\mathrm{FLRW}}_{\mu\nu}, 0)$$

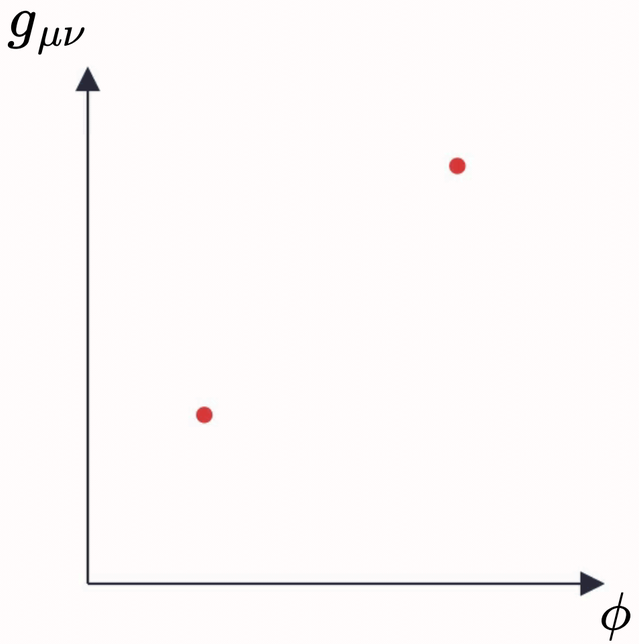


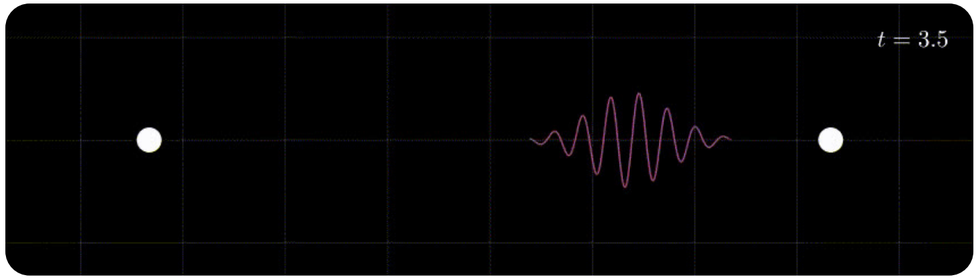
expansion of the Universe

Interpretation = choosing different pairs of (metric, scale)

$$1 + z = \frac{\nu}{\nu_0} = \frac{e^{-\phi}\,\omega[g_{\mu\nu}]\big|_e}{e^{-\phi}\,\omega[g_{\mu\nu}]\big|_o}$$

redshift





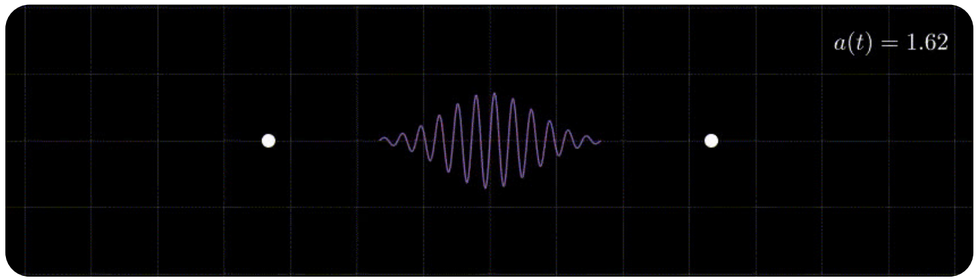
time dependent electron mass

$$(\eta_{\mu\nu}, ln(a))$$



Interpretation

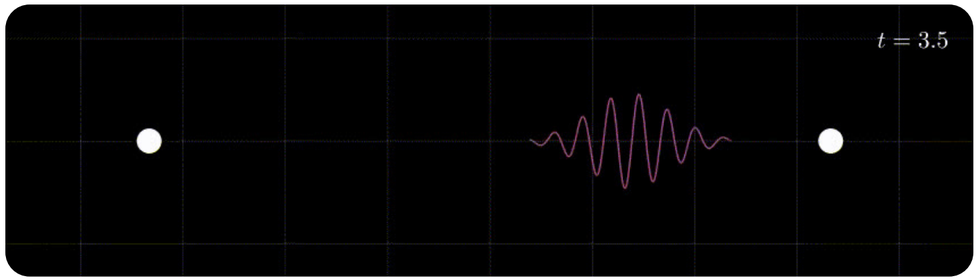
$$(g_{\mu\nu}^{\mathrm{FLRW}}, 0)$$



expansion of the Universe

$$1 + z = \frac{\nu}{\nu_0} = \frac{e^{-\phi}\,\omega[g_{\mu\nu}]\big|_e}{e^{-\phi}\,\omega[g_{\mu\nu}]\big|_o}$$

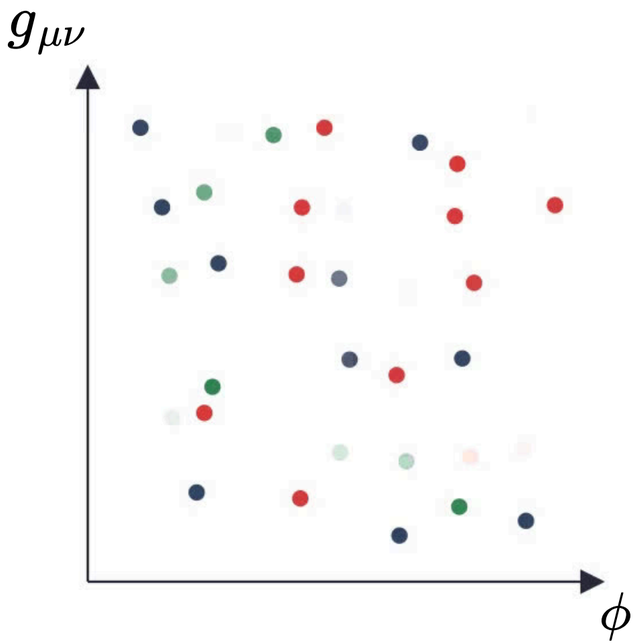
redshift



$$(\eta_{\mu\nu}, ln(a))$$

time dependent electron mass

Interpretation = choosing different pairs of (metric, scale)



Theories



Observables

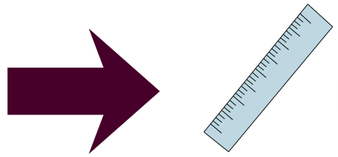
PhysRevD.65.124013

Partial: A physical quantity to which we associate a (measuring) procedure.

mass

Complete: A quantity whose value (or probability distribution) can be predicted by the theory.

redshift



independent = “interpretation invariant”

$$\iota : \mathcal{A}_{\mathrm{D}}^{0} \mapsto \mathcal{A}_{\mathrm{D}}^{\mathrm{ext}}$$

$$\mathrm{H}_T \equiv C_{\perp}[N] + C_{\parallel}[\xi] + C_W[\lambda] = \hat{C}_{\perp}[\hat{N}] + \hat{C}_{\parallel}[\hat{\xi}] + \hat{C}_W[\hat{\lambda}]$$

$$\mathrm{H}_T \equiv N\left(-\frac{p_a^2}{12M_{\mathrm{Pl}}^2 a} + \frac{p_\chi^2}{2M_{\mathrm{Pl}}^2 a^3} + M_{\mathrm{Pl}}^4 a^3 V(\chi) + M_{\mathrm{Pl}} n_0 \mu(\chi)\right) + \lambda(a p_a - p_\sigma)$$

$\mathbb{R} \times \mathbb{T}^3$

Wrap up

Outlook

1. What happens when going quantum?
2. Generalizing to the lightcone.

Conclusions

1. Scale invariance of complete observables.
2. Choice of scale equals choice of interpretation.

Thank you